

Generating trading rules on the stock markets with genetic programming

Jean-Yves Potvin^{a,c}, Patrick Soriano^{a,b}, Maxime Vallee^b

^a Centre de Recherche sur les Transports, Université de Montréal, C.P.
6128, succ. Centre-ville, Montréal, Québec, Canada H3C 3J7

^b Department d'Informatique et de Recherche Opérationnelle, Université
de Montréal, C.P. 6128, succ. Centre-ville, Montréal, Québec, Canada H3C
3J7

^c École des hautes Études Commerciales, 3000 Chemin de la Côte-Sainte-
Catherine, Montréal, Québec, Canada H3T 2A7

Computers & Operations Research, Vol. 31(2004), PP. 1033-1047

Presenter: Hung-Hsi Chen
Date: Sept. 15, 2009

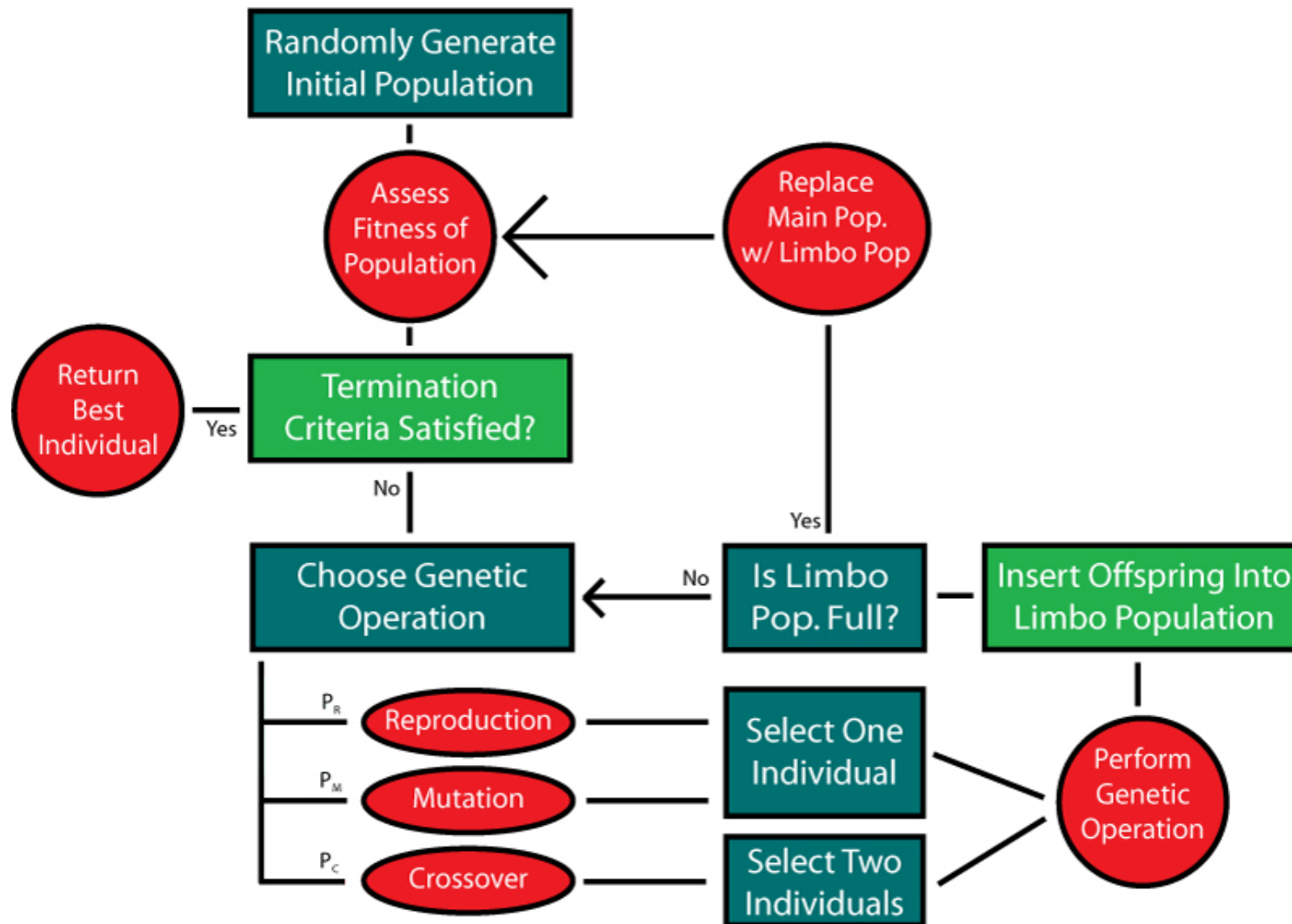
Abstract

Technical analysis is aimed at devising trading rules capable of exploiting short-term fluctuations on the financial markets. Recent results indicate that this market timing approach may be a viable alternative to the buy-and-hold approach, where the assets are kept over a relatively long time period.

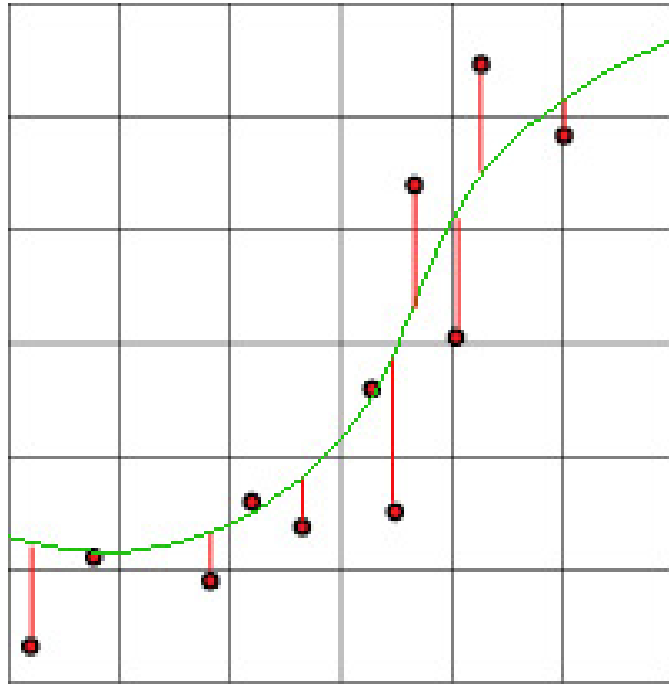
In this paper, we propose genetic programming as a means to automatically generate such short-term trading rules on the stock markets. Rather than using a composite stock index for this purpose, the trading rules are adjusted to individual stocks. Computational results, based on historical pricing and transaction volume data, are reported for 14 Canadian companies listed on the Toronto stock exchange market.

Genetic programming

- Genetic programming is a branch of genetic algorithms.
- The [main difference](#) between genetic programming and genetic algorithms is [the representation of the solution](#).



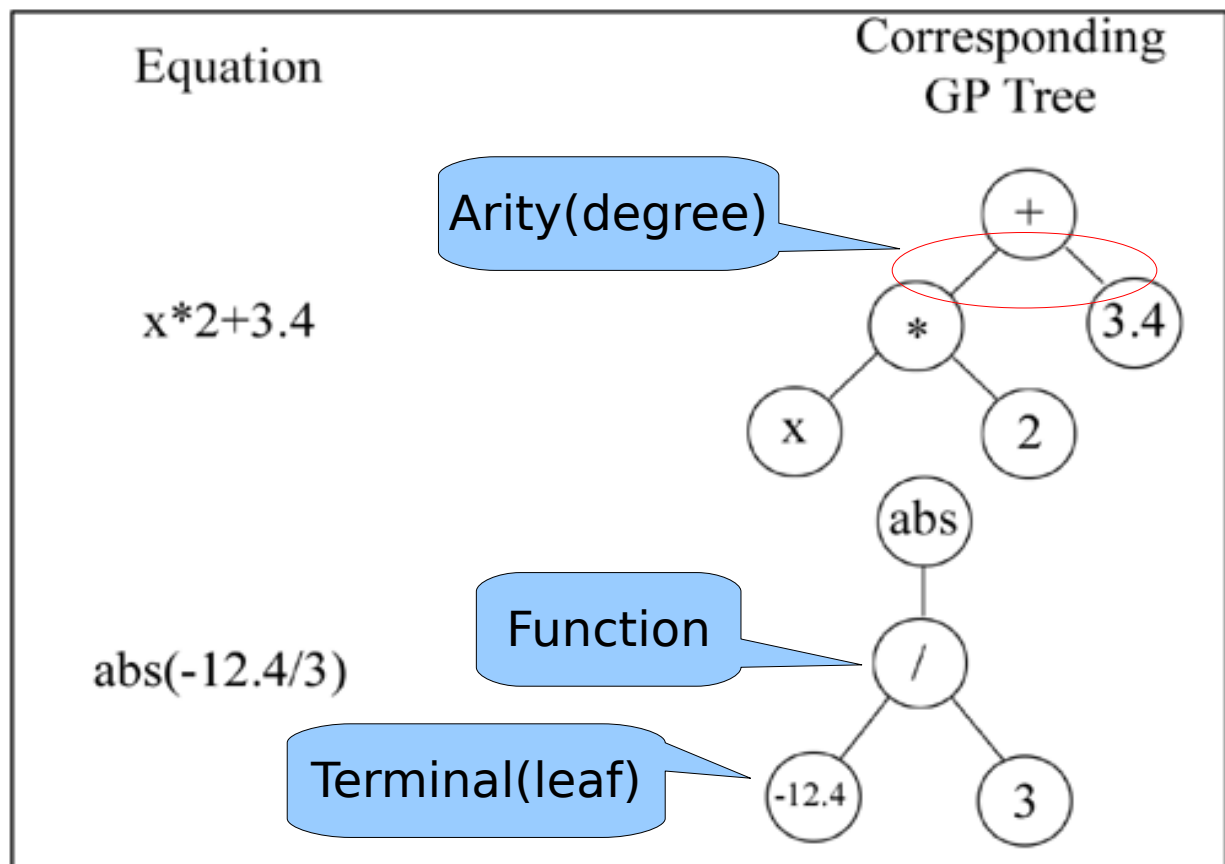
Example : symbolic Regression



Symbolic Regression is the process of determining an equation from a given set of points.

Functions = $\{+, -, *, /\}$
Terminals = $\{\text{constants, variable}\}$

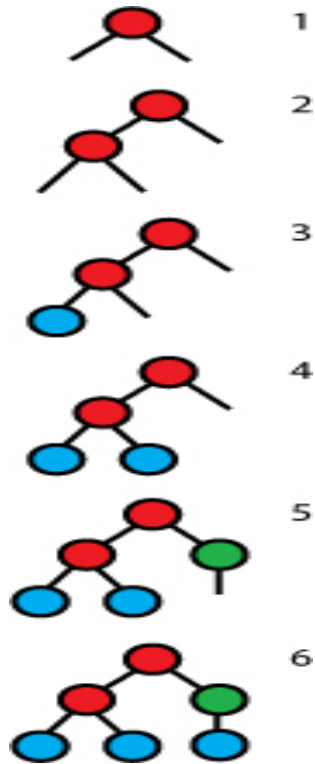
Fitness : minimize the sum of square errors



Initial population

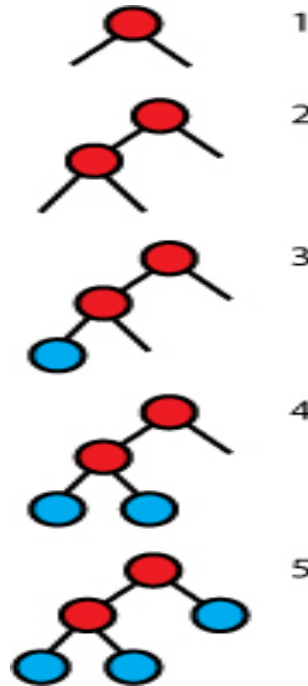
- We have determined the **MAX DEPTH**

1 、 Full method



- The length of every path between a terminal and the root is equal to a predefined depth

2 、 Grow method



- The length of a path between a terminal and the root is **at most** a predefined maximum depth.

Usually, an equal mix of trees is produced with the Full and Grow methods

Selection

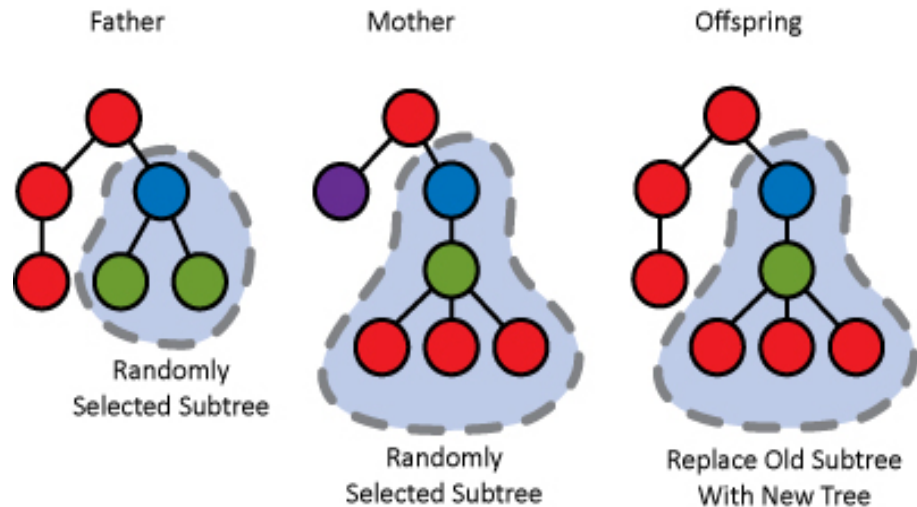
Fitness-proportionate selection scheme

$$prob_i = \frac{fitness_i}{\sum_{j=1}^P fitness_j} \quad , P \text{ is the size of population}$$

Reproduction

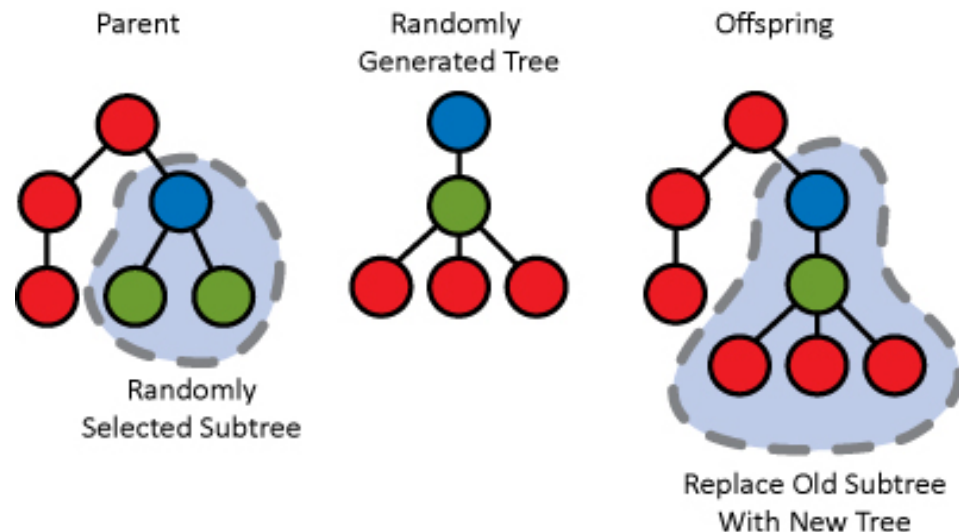
- Reproduction simply makes a copy of the selected program in the new population, without any modification.
- The **goal** of reproduction is to **increase the prevalence of individuals which have proven themselves fit to solve the problem.**

Crossover



The probability distribution for the selection of a crossover point favors the exchange of subtrees rather than terminals.

Mutation



- First, a random point is chosen in the tree.
- This point, as well as the subtree below it, are then removed and replaced by a new, randomly generated subtree.

Generating trading rules with GP

- These rules are boolean functions which return either a buy(True) or a sell signal(False)

Functions :

- Arithmetic operators: $+$, $-$, $*$, $/$
- Boolean operators: and, or, not
- Relational operators: $<$, $>$
- Boolean functions: if-then-else

below **variable s** stands either for constant **Price or Volume**

- $\text{norm}(r1, r2)$: absolute value of $(r1-r2)$
- $\text{avg}(s, n)$: average s over the past n days
- $\text{max}(s, n)$: maximum value of s over the past n days
- $\text{min}(s, n)$: minimum value of s over the past n days
- $\text{lag}(s, n)$: s is lagged by n days
- $\text{volatility}(n)$: variance in daily returns over the past n days
- $\text{RSI}(n)$: relative strength index
- $\text{ROC}(n)$: rate of change

$$RSI(n) = 100 - \left(\frac{100}{1 + RS(n)} \right), \quad \text{where } RS(n) = \frac{\sum_{i \in D^+(n)} r_i}{-\sum_{i \in D^-(n)} r_i}$$

$$ROC(n) = \left(\frac{\text{closing price of current day}}{\text{closing price } n \text{ days ago}} - 1 \right) \times 100$$

Terminals :

Constants

Real : chosen in the interval [0, 250]

Boolean : True, False

Others : Price, Volume

Real variables :

p: stock price of the current day

v : transaction volume of the current day

- Real functions are always found in the lower part of the tree
- boolean functions and operators in the upper part(including the root)
- Relational operators make the transition between the two

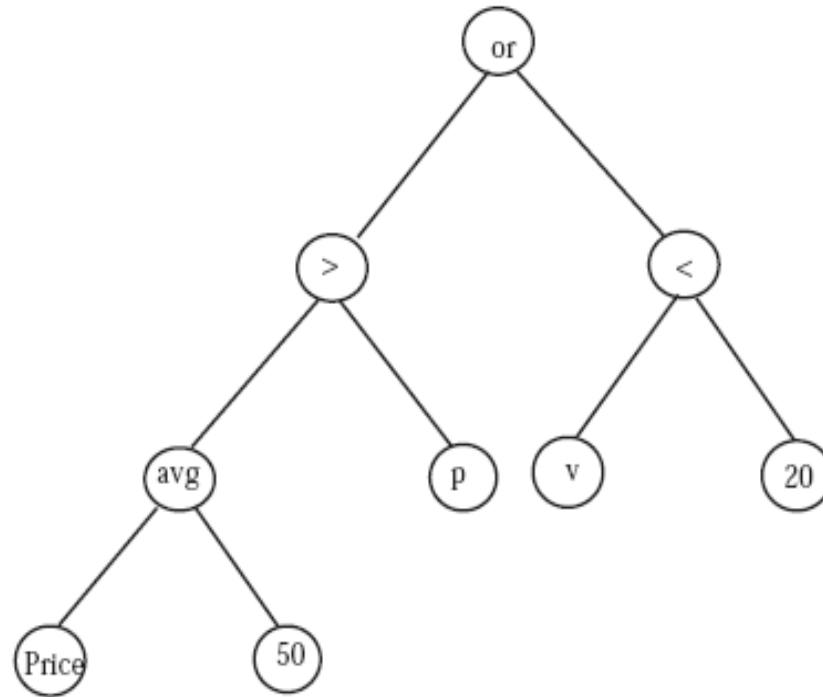


Fig. 4. Example of a trading rule.

Buy : average stock price over the past 50 days is greater then the current price or the current transaction volume is less than 20

Sell : otherwise

Initial population

- Mix of trees produced by the full and grow methods.

Tree construction guidelines :

- The root of the tree is selected among the boolean functions and operators
- The descendants of root may be selected among boolean constants, boolean functions, boolean or relational operators.
- When a relational operator is selected, its descendants must be selected among real functions or terminals.

Fitness evaluation

raw fitness value : the excess return of a trading rule over the buy-and-hold approach, as evaluated on historical data

Selection

First, all programs in the current population are sorted from best(rank 1) to worst(rank P) based on their raw fitness value. Then use **Stochastic universal sampling** for selection

**Adjusted fitness value
with rank i :**

$$f_i = Max - \left[(Max - Min) \frac{i - 1}{P - 1} \right]$$

Ex :

raw fitness value = [1, 4, 2, 3, **12**], prob = [4.5%, 18%, 9%, 14%, **54.5%**]

Adjusted fitness value = [1, 9.25, 3.75, 6.5, **12**] with equal distance
prob = [3%, 28.5%, 11.5%, 20%, **37%**]

Reproduction

- Same as previous introduction
- Reproduction simply makes a copy of the selected program in the new population, without any modification.

Crossover

- Subtree crossover
- Once a cross point is chosen on the first parent, the subset of compatible cross points is identified on the other parent and only this subset is considered for crossover.
- If there is no compatible cross point, two other parents are selected.

Experimental results

- Stocks of TSE 300 index(Toronto Stock Exchange, since 1852)
- 14 different activity sectors
- One company was selected in each sector, but only among those which were active on the market before 1990.

Table 1
Canadian companies in the study

Activity sector	Company	Symbol
Mining	Alcan Aluminium Ltd	<i>AL</i>
Precious metals	Barrick Gold Corporation	<i>ABX</i>
Oil and gas	Cdn. Occidental Petroleum Ltd	<i>CXY</i>
Forestry 森林業	Abitibi-Consolidated Inc	<i>A</i>
Consumer products	Molson Inc	<i>MOL</i>
Industrial products	Bombardier Inc	<i>BBD</i>
Real estate 不動業	Cambridge Shopping Centers	<i>CBG</i>
Transportation and env.	Trimac Corporation	<i>TMA</i>
Pipelines	TransCanada Pipelines Ltd	<i>TRP</i>
Utilities	Canadian Utilities Ltd	<i>CU</i>
Communications	The Seagram Company Ltd	<i>VO</i>
Merchandising 行銷業	Sears Canada Inc	<i>SCC</i>
Financial services	Royal Bank of Canada	<i>RY</i>
Diversified (conglomerate)	Canadian Pacific Ltd	<i>CP</i>

Short training period:

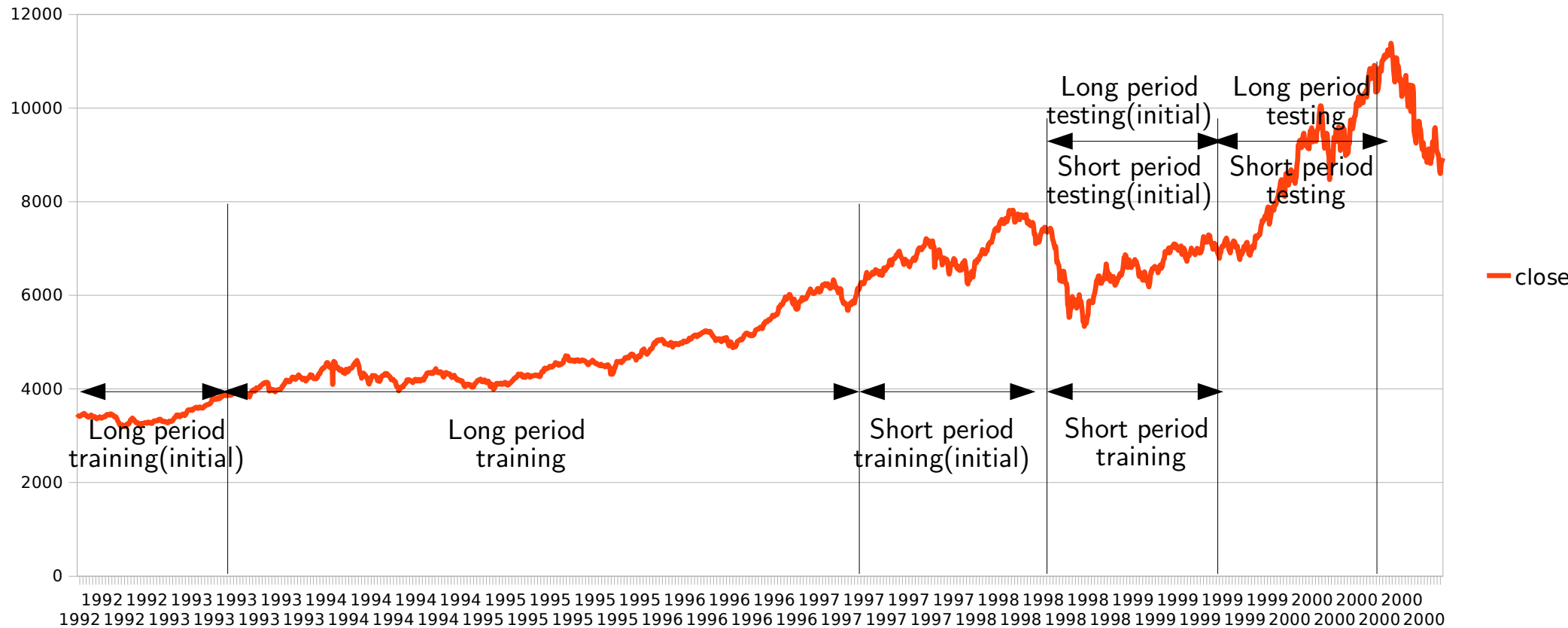
for function $\text{avg}(s, 250)$

Initial (for training):	June 25, 1997 to June 22, 1998 (250 days),
Training:	June 22, 1998 to June 25, 1999 (256 days),
Initial (for testing):	July 07, 1998 to June 25, 1999 (250 days),
Testing:	June 28, 1999 to June 30, 2000 (256 days).

Long training period:

Initial (for training):	June 30, 1992 to July 02, 1993 (250 days),
Training:	July 02, 1993 to June 25, 1999 (1498 days),
Initial (for testing):	July 07, 1998 to June 25, 1999 (250 days),
Testing:	June 28, 1999 to June 30, 2000 (256 days).

TSE300 index



Parameter settings	
Population size	500
Number of generations	50
Initialization method	Equal mix of Full and Grow
Selection method	Ranking + SUS
Reproduction rate	0.35
Crossover rate	0.60
Mutation rate	0.05
Probability of choosing an internal point (crossover)	0.90
Probability of choosing a terminal (crossover)	0.10
Maximum depth of programs in initial population	6
Maximum depth of programs in following generations	17

Numerical results for the short training period [average of 10 runs](#)

Symbol	256 days	Training	256 days	Testing
<i>A</i>		144.16%		15.29%
<i>ABX</i>		202.16%		38.12%
<i>AL</i>		135.95%		15.79%
<i>BBD</i>		123.04%		−34.92%
<i>CBG</i>		174.01%		−75.09%
<i>CP</i>		120.70%		0.86%
<i>CU</i>		56.69%		7.88%
<i>CXY</i>		183.40%		−102.16%
<i>MOL</i>		99.23%		17.38%
<i>RY</i>		125.68%		3.25%
<i>SCC</i>		110.80%		−21.84%
<i>TMA</i>		211.93%		42.27%
<i>TRP</i>		85.78%		36.04%
<i>VO</i>		146.78%		−10.83%
Average		131.17%		−4.85%
Number of positive returns		14		9

Numerical results for the long training period average of 10 runs

Symbol	1498 days Training	256 days Testing
<i>A</i>	31.00%	18.55%
<i>ABX</i>	34.27%	7.69%
<i>AL</i>	21.99%	19.83%
<i>BBD</i>	18.30%	−79.46%
<i>CBG</i>	33.15%	−35.58%
<i>CP</i>	19.67%	12.11%
<i>CU</i>	9.77%	6.45%
<i>CXY</i>	30.39%	−44.85%
<i>MOL</i>	19.36%	13.74%
<i>RY</i>	19.59%	8.13%
<i>SCC</i>	16.53%	−7.33%
<i>TMA</i>	30.57%	54.80%
<i>TRP</i>	16.57%	21.09%
<i>VO</i>	19.88%	−45.41%
Average	22.93%	−3.59%
Number of positive returns	14	9

Evolution of the trading rules over the long training period

Symbol	Generation	Initial population		Final population	
		Best (%)	Average (%)	Best (%)	Average (%)
<i>A</i>	37	15.08	−3.53	27.48	21.89
<i>ABX</i>	42	20.76	3.28	31.76	27.13
<i>AL</i>	36	8.53	−10.32	20.03	14.93
<i>BBD</i>	36	3.50	−37.89	14.80	8.53
<i>CBG</i>	37	24.73	10.86	31.90	28.38
<i>CP</i>	34	7.34	−8.80	17.36	12.13
<i>CU</i>	37	2.57	−10.21	8.72	5.16
<i>CXY</i>	37	18.02	−7.09	29.23	24.28
<i>MOL</i>	38	9.73	−2.44	17.51	13.61
<i>RY</i>	33	8.70	−14.91	17.85	12.44
<i>SCC</i>	40	4.90	−24.99	14.50	9.66
<i>TMA</i>	37	18.69	2.26	29.17	24.09
<i>TRP</i>	39	7.41	−3.73	14.62	11.45
<i>VO</i>	40	7.09	−13.42	17.78	12.94
Average	37	11.22	−8.64	22.93	16.19

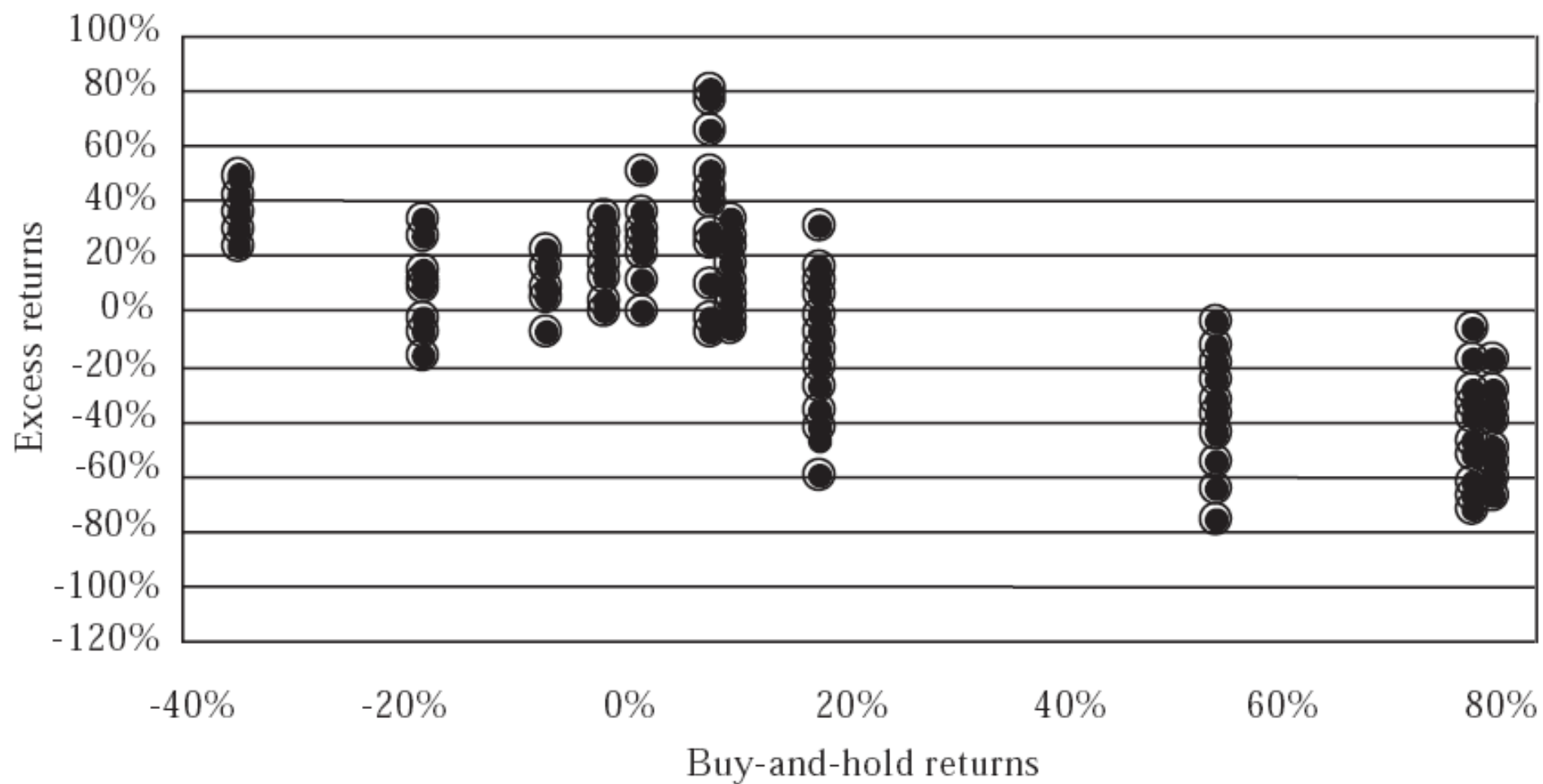


Fig. 5. Excess returns versus buy-and-hold returns on the testing period.

Thank you!!